

MTH 446: Lecture # 11

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Lecture Span

1. CR equations

Cauchy Riemann Equations

The derivative w.r.t the real number line is the same as that on the complex number line. Note that $z = z_0 + h$, then

$$x + iy = (x_0 + h) + iy_0 \quad (1)$$

Therefore,

$$\frac{f(z) - f(z_0)}{z - z_0} = \frac{(u(x + h, y) - u(x, y)) + i(v(x + h, y) - v(x, y))}{h} \quad (2)$$

$$= \partial_x u + i\partial_x v \quad (3)$$

Similarly, taking the partial derivative on the complex plane yields:

$$\frac{f(z) - f(z_0)}{z - z_0} = \frac{(u(x, y + h) - u(x, y)) + i(v(x, y + h) - v(x, y))}{ih} \quad (4)$$

$$= -i\partial_y u + \partial_y v \quad (5)$$

Therefore,

$$\partial_x u = \partial_y v, \quad \partial_x v = -\partial_y u \quad (6)$$

Example

Let $f(z) = x + iy$. Here, the CR equations are not satisfied. Therefore, $f'(z)$ DNE for all $z \in \mathbb{C}$.

Example

Consider $f(z) = x^2 + y^2$. $f'(z)$ DNE if $z \neq 0$ by the necessary conditions of complex differentiability. next,

$$\lim_{z \rightarrow 0} \frac{|z|^2}{z} = \lim \bar{z} = 0 \quad (7)$$

Therefore, f is complex differentiable only at the origin.

Example

$f(z) = \bar{z} \operatorname{Im}(z)$. Here,

$$f(z) = xy - iy^2 \quad (8)$$

Here, we obtain that $f(z)$ is not complex differentiable for $z \neq 0$ by the necessary condition of complex differentiability.

At $f'(z)$,

$$f'(z) = \lim_{z \rightarrow 0} \frac{\bar{z} \operatorname{Im}(z)}{z} = \lim_{z \rightarrow 0} \frac{|\bar{z}| |\operatorname{Im}(z)|}{|z|} = \lim_{z \rightarrow 0} |y| = 0 \quad (9)$$

Therefore, f is complex differentiable only at the origin.

Example

For $f(z) = z^2 = (x^2 + y^2) + 2xyi$. The C.R. equations are satisfied at every point, but we cannot conclude that $d/dz(z^2)$ exists at every point $z \in \mathbb{C}$. We need the next theorem...

Theorem 0.1. *Sufficient conditions for complex differentiability. Let $f : G \rightarrow \mathbb{C}$ be a function. If u_x, u_y, v_x, v_y are continuous at $z = z_0$ and the CR equations are satisfied at $z = z_0$. Then $f'(z_0)$ exists.*

Example

Use the previous example $f(z) = z^2$. Then, the CR equations are satisfied everywhere, and thus f is CD everywhere on $\mathbb{C}$.

Must verify that v_i and u_i are continuous!!!

Example

$f(z) = 2x + ixy$.

Formal ending: "By the necessary condition of complex differentiability, $f'(z)$ does not exist if $z \neq 2$ ".